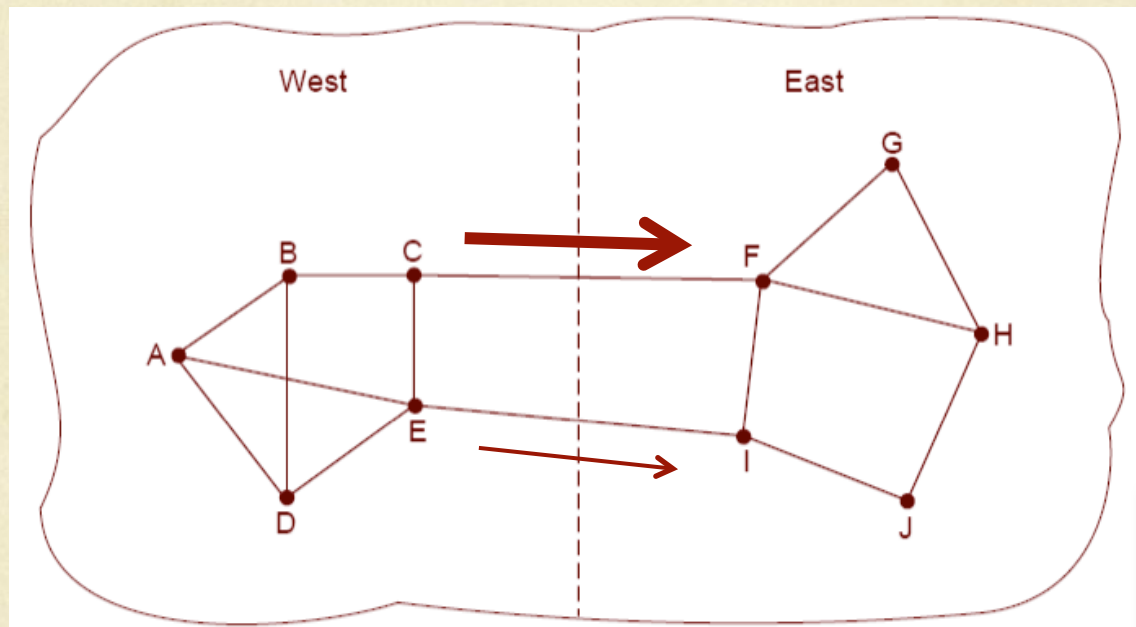


Congestion control

- ◆ Responsibility shared by *transport* & *network* layers
 - ◆ We discuss network layer aspects here
- ◆ Several approaches could be taken to handle congestion
 - ◆ Let routing algorithm handle it (*traffic-aware routing*)
 - ◆ Decrease traffic/load
 - ◆ Increase capacity
 - ◆ Add more resources to network (*provisioning*)
 - ◆ *More of a long-term solution*

Traffic-aware routing

- ◆ *Goal*: consider *load* & direct traffic away from *hotspots*
- ◆ Choose routes depending on traffic, not just topology



- ◆ I.e., use EI for West-to-East traffic if CF is loaded
- ◆ Could lead to *oscillations* in routing tables

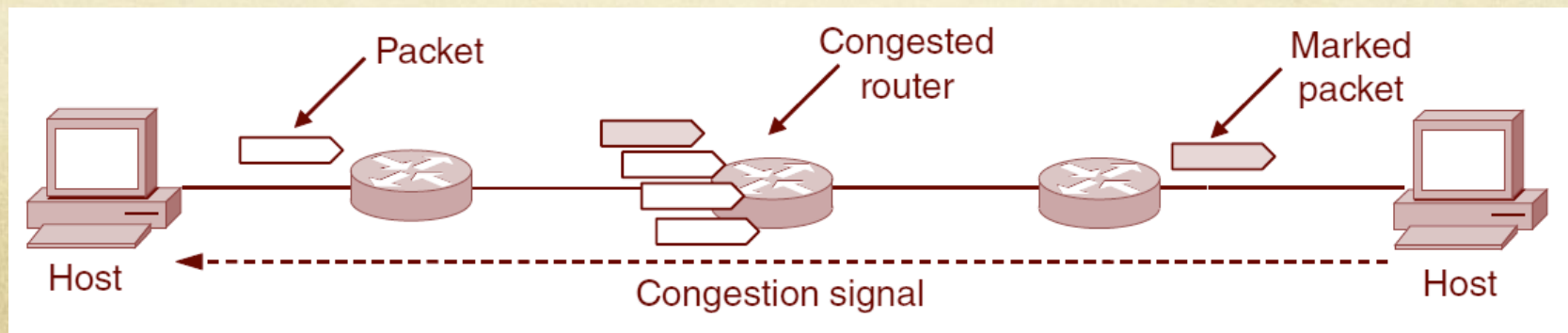
Just routing all traffic in another direction could cause that new direction to get overloaded!

Traffic-aware routing

- ◆ Solutions to oscillation problem
 - ◆ Change routes slowly, not all at once
 - ◆ Use external *traffic engineering* to decide on weights, etc.
 - ◆ Use multiple paths at once
 - ◆ I.e., *distribute* traffic over links (e.g., use both CF & EI)

Traffic throttling

- ◆ Routers *determine* when congestion is impending
- ◆ Routers *notify* senders of impending congestion
- ◆ Some approaches
 - ◆ Send *choke packet* to senders causing congestion
 - ◆ Sender required to reduce rate of transmission
 - ◆ Set flag in packets notifying impending congestion
 - ◆ Explicit congestion notification (*ECN*)



Admission control

- ◆ Useful in *connection-oriented* service
- ◆ Refuse new *connection* (virtual circuit) if network can handle it without being congested
- ◆ How do you know if new connection will cause congestion?
 - ◆ Need some way to represent traffic state
 - ◆ Approaches include *leaky bucket*

And if nothing else works...

- ◆ ...routers resort to *load shedding*
- ◆ I.e., routers simply *drop* packets when inundated
- ◆ *Which* packets should be dropped?
 - ◆ Depends...
 - ◆ File transfer → *old* packet worth more than *new* one
 - ◆ Real-time streaming → *new* packet worth more than *old* one